ORF 309 Midterm 1 Cheat Sheet

Slides 01–09 | Through Section 2.3

Probability Foundations

Sample space Ω: set of all outcomes. **Event**: $A \subseteq \Omega$.
Set ops: $A \cup B$, $A \cap B$, A^c, $A \setminus B = A \cap B^c$.
De Morgan: $(A \cup B)^c = A^c \cap B^c$; $(A \cap B)^c = A^c \cup B^c$.
Partition: $B_1, \ldots, B_n$ disjoint, $\bigcup B_i = \Omega$.

Kolmogorov Axioms

(1) $P(A) \geq 0$ (2) $P(\Omega) = 1$ (3) If A_i pairwise disjoint: $P(\bigcup A_i) = \sum P(A_i)$
Consequences: $P(\emptyset) = 0$; $P(A^c) = 1 - P(A)$; $A \subseteq B \Rightarrow P(A) \leq P(B)$
$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Inclusion–Exclusion: $P(\bigcup_{i=1}^n A_i) = \sum_i P(A_i) - \sum_{i<j} P(A_i \cap A_j) + \cdots$
Union Bound: $P(\bigcup A_i) \leq \sum P(A_i)$
Continuity: $A_n \uparrow A \Rightarrow P(A_n) \to P(A)$; same for $\downarrow$.

Counting

Equally likely: $P(A) = |A|/|\Omega|$
Multiplication rule: k stages with n_i choices $\Rightarrow \prod n_i$ total.
Permutations (ordered, no repl): $P(n,k) = \frac{n!}{(n-k)!}$
Combinations (unordered, no repl): $\binom{n}{k} = \frac{n!}{k!(n-k)!}$
Multinomial: $\binom{n}{k_1,\ldots,k_r} = \frac{n!}{k_1! \cdots k_r!}$, $\sum k_i = n$
Stars & bars: n balls into k bins: $\binom{n+k-1}{k-1}$
Binomial thm: $(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}$
$\binom{n}{k} = \binom{n}{n-k}$; $\sum_{k=0}^n \binom{n}{k} = 2^n$
Vandermonde: $\binom{m+n}{k} = \sum_j \binom{m}{j}\binom{n}{k-j}$
Derangements: $D_n = n! \sum_{k=0}^n \frac{(-1)^k}{k!}$
Exactly k fixed pts: $\binom{n}{k} \cdot D_{n-k}$
Lattice paths $(0,0) \to (m,n)$: $\binom{m+n}{n}$
Through (a,b): $\binom{a+b}{b} \cdot \binom{(m-a)+(n-b)}{n-b}$

Conditional Probability

$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$, $P(B) > 0$
Multiplication: $P(A \cap B) = P(A \mid B)\, P(B)$
Chain rule: $P(\bigcap_{i=1}^n A_i) = P(A_1) \prod_{i=2}^n P(A_i \mid \bigcap_{j<i} A_j)$
Law of Total Probability (LOTP): $P(A) = \sum_i P(A \mid B_i)\, P(B_i)$ (B_i partition Ω)
Two-event: $P(A) = P(A \mid B)P(B) + P(A \mid B^c)P(B^c)$
Bayes' Rule: $P(B_j \mid A) = \frac{P(A|B_j)\, P(B_j)}{\sum_i P(A|B_i)\, P(B_i)}$
Prior $P(B_i) \to$ *Posterior* $P(B_i \mid A)$ via *likelihood* $P(A \mid B_i)$.
Sequential Bayes: Use posterior as new prior for next observation.

Independence

$A \perp B \iff P(A \cap B) = P(A)P(B)$
Equiv: $P(A \mid B) = P(A)$ when $P(B) > 0$.
$A \perp B \Rightarrow A \perp B^c$, $A^c \perp B$, $A^c \perp B^c$.
Mutually exclusive $\neq$ independent! If $A \cap B = \emptyset$ and $P(A), P(B) > 0$: A, B are **dependent**.
Mutual independence: For *every* subset $S \subseteq \{1,\ldots,n\}$, $|S| \geq 2$: $P(\bigcap_{i \in S} A_i) = \prod_{i \in S} P(A_i)$
Pairwise $\not\Rightarrow$ mutual independence.
Cond. indep: $A \perp B \mid C \iff P(A \cap B \mid C) = P(A \mid C)P(B \mid C)$
Does **not** imply unconditional independence (or vice versa).

Discrete Random Variables

RV: $X : \Omega \to \mathbb{R}$.
PMF: $p_X(x) = P(X = x)$. $p_X(x) \geq 0$, $\sum_x p_X(x) = 1$.
CDF: $F_X(x) = P(X \leq x)$.
Non-decreasing, right-continuous, $F(-\infty) = 0$, $F(\infty) = 1$.
$P(a < X \leq b) = F(b) - F(a)$; $P(X > a) = 1 - F(a)$.
Discrete: step function, jump at $x_i = p_X(x_i)$.
Functions of RVs: $Y = g(X)$: $p_Y(y) = \sum_{x: g(x)=y} p_X(x)$
Indicator: $\mathbf{1}_A = \begin{cases} 1 & \omega \in A \\ 0 & \text{else} \end{cases}$; $E[\mathbf{1}_A] = P(A)$.
Independence of RVs: $X \perp Y \iff$ for all x, y:
$P(X{=}x, Y{=}y) = P(X{=}x)\, P(Y{=}y)$

Named Discrete Distributions

Bernoulli(p)
$X \in \{0,1\}$. $P(X{=}1) = p$. $E[X] = p$, $\text{Var} = p(1-p)$.

Binomial(n,p)
$X = \#$ successes in n indep. Bern(p) trials.
$P(X{=}k) = \binom{n}{k} p^k (1-p)^{n-k}$, $k = 0, \ldots, n$
$E[X] = np$, $\text{Var} = np(1-p)$.
Sum: $\text{Bin}(n,p) + \text{Bin}(m,p) = \text{Bin}(n+m, p)$ (indep).

Geometric(p)
T_1 = trial of 1st success. $T_1 \in \{1,2,3,\ldots\}$.
$P(T_1{=}k) = (1-p)^{k-1} p$; $P(T_1 > k) = (1-p)^k$
$E[T_1] = 1/p$, $\text{Var} = (1-p)/p^2$.
Memoryless: $P(T_1 > m+n \mid T_1 > m) = P$
Only discrete memoryless distribution.

Negative Binomial(r,p)
T_r = trial of r-th success. $T_r \in \{r, r+1, \ldots\}$.
$P(T_r{=}n) = \binom{n-1}{r-1} p^r (1-p)^{n-r}$
$E[T_r] = r/p$, $\text{Var} = r(1-p)/p^2$.
T_r = sum of r i.i.d. Geom(p). NegBin$(1,p)$ =Geom(p).

Poisson(λ)
$P(X{=}k) = e^{-\lambda} \frac{\lambda^k}{k!}$, $k = 0, 1, 2, \ldots$
$E[X] = \lambda$, $\text{Var} = \lambda$.
Poisson approx: n large, p small, $\lambda = np$: $\text{Bin}(n,p) \approx \text{Pois}(\lambda)$.
Sum: $\text{Pois}(\lambda_1) + \text{Pois}(\lambda_2) = \text{Pois}(\lambda_1 + \lambda_2)$ (indep).
Thinning: Mark each w.p. p indep $\Rightarrow$ $\text{Pois}(\lambda p)$.
Splitting: Given $N(s+t) = n$: $N(s) \sim \text{Bin}(n, s/(s+t))$.

Hypergeometric
Pop N: K success, $N-K$ fail. Draw n w/o replacement.
$P(X{=}k) = \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}$; $E[X] = nK/N$

Expectation

$E[X] = \sum_x x\, p_X(x)$
LOTUS: $E[g(X)] = \sum_x g(x)\, p_X(x)$ (no need for PMF of $g(X)$)
Linearity (always, no indep needed): $E[aX + bY + c] = aE[X] + bE[Y] + c$
$X \perp Y \Rightarrow E[XY] = E[X]\, E[Y]$ (converse false)
Conditional expectation: $E[X \mid B] = \sum_x x\, P(X{=}x \mid B)$
Tower / Law of Total Expectation: $E[X] = \sum_i E[X \mid B_i]\, P(B_i)$ (B_i partition)
$E[X] = E[E[X \mid Y]]$
Tail sum formula ($X \in \{0,1,2,\ldots\}$): $E[X] = \sum_{k=1}^\infty P(X \geq k) = \sum_{k=0}^\infty P(X > k)$
Indicator trick: Write $X = \sum \mathbf{1}_{A_i}$, then $E[$
Works even if dependent!

Variance

$\text{Var}(X) = E[(X-\mu)^2] = E[X^2] - (E[X])^2$
$\text{SD}(X) = \sqrt{\text{Var}(X)}$
$\text{Var}(X) \geq 0$; $= 0$ iff X is constant a.s.
$\text{Var}(aX + b) = a^2 \text{Var}(X)$
$E[X^2] = \text{Var}(X) + (E[X])^2$ (**key identity!**)
$X \perp Y$: $\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y)$
General: $\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X,Y)$
$\text{Var}(\sum_i X_i) = \sum_i \text{Var}(X_i) + 2\sum_{i<j} \text{Cov}(X_i, X_j)$
Bin var via indicators: $X = \sum I_i$ (indep Bern),

$\mathrm{Var}(X) = n\,p(1-p)$.

Covariance & Correlation

$\mathrm{Cov}(X,Y) = E[XY] - E[X]E[Y]$

$\mathrm{Cov}(X,X) = \mathrm{Var}(X)$; $\mathrm{Cov}(X,Y) = \mathrm{Cov}(Y,X)$

$\mathrm{Cov}(aX+b, cY+d) = ac\,\mathrm{Cov}(X,Y)$

$\mathrm{Cov}(X+Y,Z) = \mathrm{Cov}(X,Z) + \mathrm{Cov}(Y,Z)$ (bilinear)

$X \perp Y \Rightarrow \mathrm{Cov}(X,Y) = 0$ (converse false)

$\rho(X,Y) = \frac{\mathrm{Cov}(X,Y)}{\mathrm{SD}(X)\,\mathrm{SD}(Y)}$; $-1 \le \rho \le 1$

$|\rho| = 1 \iff Y = aX + b$ (perfect linear).

Uncorrelated ($\rho = 0$) $\neq$ independent.

Var of linear combo:

$\mathrm{Var}(aX+bY) = a^2\mathrm{Var}(X) + b^2\mathrm{Var}(Y) + 2a$

Moment Generating Functions

$M_X(t) = E[e^{tX}]$ (defined near $t=0$)

$E[X^k] = M_X^{(k)}(0)$ (k-th derivative at 0)

MGF uniquely determines distribution.

$M_{aX+b}(t) = e^{bt}M_X(at)$

$X \perp Y$: $M_{X+Y}(t) = M_X(t)\cdot M_Y(t)$

MGFs: Bern(p): $1-p+pe^t$. Bin(n,p): $(1-p+pe^t)^n$.

Geom(p): $\frac{pe^t}{1-(1-p)e^t}$. Pois(λ): $e^{\lambda(e^t-1)}$.

Continuous Random Variables

PDF: $f_X(x) \ge 0$, $\int_{-\infty}^{\infty} f_X(x)\,dx = 1$

$P(a \le X \le b) = \int_a^b f_X(x)\,dx$; $P(X = a) = 0$

CDF: $F_X(x) = \int_{-\infty}^{x} f_X(t)\,dt$; $f_X(x) = F_X'(x)$

Expectation: $E[X] = \int x\,f_X(x)\,dx$

LOTUS: $E[g(X)] = \int g(x)\,f_X(x)\,dx$

Variance: $\mathrm{Var}(X) = \int (x-\mu)^2 f_X(x)\,dx = E[X^2] - (E[X])^2$

Uniform $U(a,b)$

$f(x) = \frac{1}{b-a}$ for $a \le x \le b$, else 0.

$E[X] = \frac{a+b}{2}$, $\mathrm{Var} = \frac{(b-a)^2}{12}$.

Exponential $\mathrm{Exp}(\lambda)$

$f(x) = \lambda e^{-\lambda x}$, $x \ge 0$. $F(x) = 1 - e^{-\lambda x}$.

$P(X > t) = e^{-\lambda t}$.

$E[X] = 1/\lambda$, $\mathrm{Var} = 1/\lambda^2$.

Memoryless: $P(X > s+t \mid X > s) = P(X > t)$

Only continuous memoryless distribution.

MGF: $M(t) = \lambda/(\lambda - t)$, $t < \lambda$.

Normal $N(\mu, \sigma^2)$

$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}$

$E[X] = \mu$, $\mathrm{Var} = \sigma^2$.

Standardize: $Z = \frac{X-\mu}{\sigma} \sim N(0,1)$.

$P(a < X < b) = \Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right)$

$\Phi(-z) = 1 - \Phi(z)$. $P(|Z| \le z) = 2\Phi(z) - 1$.

$aX + b \sim N(a\mu + b, a^2\sigma^2)$.

Sum: $X \perp Y$, $X \sim N(\mu_1, \sigma_1^2)$, $Y \sim N(\mu_2, \sigma_2^2)$:

$X + Y \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$.

MGF: $M(t) = \exp(\mu t + \sigma^2 t^2/2)$.

Functions of Continuous RVs

CDF method: $F_Y(y) = P(g(X) \le y)$, then differentiate.

Monotone transform: If $Y = g(X)$, g strictly monotone:

$f_Y(y) = f_X(g^{-1}(y)) \cdot |(g^{-1})'(y)|$

Quantiles: q-th quantile x_q: $F(x_q) = q$. Median: $q = 0.5$.

Joint Distributions (Discrete)

Joint PMF: $p_{X,Y}(x,y) = P(X{=}x, Y{=}y)$

Marginals: $p_X(x) = \sum_y p_{X,Y}(x,y)$; $p_Y(y) = \sum_x p_{X,Y}(x,y)$

Conditional: $p_{X|Y}(x|y) = p_{X,Y}(x,y)/p_Y(y)$

Independence: $p_{X,Y}(x,y) = p_X(x)\,p_Y(y)$ for all x, y.

$E[g(X,Y)] = \sum_x \sum_y g(x,y)\,p_{X,Y}(x,y)$

Convolution (discrete):

$P(X{+}Y{=}k) = \sum_j P(X{=}j)\,P(Y{=}k{-}j)$

Multinomial: n trials, k outcomes w/ probs $p_1, \ldots, p_k$:

$P(X_1{=}n_1, \ldots, X_k{=}n_k) = \frac{n!}{n_1!\cdots n_k!} p_1^{n_1} \cdots p_k^{n_k}$

Marginals of multinomial are binomial.

Joint Distributions (Continuous)

Joint PDF: $f_{X,Y}(x,y) \ge 0$, $\iint f\,dx\,dy = 1$

$P((X,Y) \in A) = \iint_A f_{X,Y}(x,y)\,dx\,dy$

Marginals: $f_X(x) = \int f_{X,Y}(x,y)\,dy$

Conditional: $f_{X|Y}(x|y) = f_{X,Y}(x,y)/f_Y(y)$

Independence: $f_{X,Y}(x,y) = f_X(x)\,f_Y(y)$ for all x, y.

Factorization test: If $f_{X,Y}(x,y) = g(x)\,h(y)$ over a *rectangular* domain, then $X \perp Y$.

Convolution (continuous):

$f_{X+Y}(z) = \int f_X(x)\,f_Y(z-x)\,dx$

$E[g(X,Y)] = \iint g(x,y)\,f_{X,Y}(x,y)\,dx\,dy$

Inequalities & Limit Theorems

Markov: $X \ge 0$: $P(X \ge a) \le E[X]/a$

Chebyshev: $P(|X - \mu| \ge a) \le \mathrm{Var}(X)/a^2$

Weak LLN: i.i.d. X_i, $E[X_i] = \mu$, $\mathrm{Var}(X_i) = \sigma^2$:

$\bar{X}_n = \frac{1}{n}\sum X_i \xrightarrow{P} \mu$

Proof: $P(|\bar{X}_n - \mu| \ge \epsilon) \le \frac{\sigma^2}{n\epsilon^2} \to 0$.

Useful Series

$\sum_{k=0}^{\infty} r^k = \frac{1}{1-r}$, $|r| < 1$

$\sum_{k=1}^{\infty} k r^{k-1} = \frac{1}{(1-r)^2}$

$\sum_{k=0}^{\infty} k r^k = \frac{r}{(1-r)^2}$ $\sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = e^{\lambda}$

$\sum_{k=0}^{n} \binom{n}{k} = 2^n$

Problem-Solving Techniques

1. Indicator trick: $X = \sum \mathbf{1}_{A_i} \Rightarrow E[X] = \sum P(A_i)$. For Var, need independence.

2. LOTP: Partition and sum $P(A|B_i)P(B_i)$.

3. First-step analysis: Condition on 1st step, write recursion for $P(A)$, solve.

4. Tail sum: $E[X] = \sum_{k \ge 1} P(X \ge k)$ for $X \in \{0, 1, \ldots\}$.

5. $E[X^2] = \mathrm{Var} + \mu^2$: Always use this to recover second moments.

6. Tower property: $E[X] = E[E[X|Y]]$ to decompose by cases.

7. MGF method: Multiply MGFs of indep RVs, recognize distribution.

8. Poisson approx: n large, p small $\Rightarrow$ Bin(n,p) $\approx$Pois(np).

9. Sequential Bayes: Use posterior as new prior.

10. Complementary counting: $P(A) = 1 - P(A^c)$.

Common Exam Traps

- Linearity of E needs NO independence; Var of sum DOES.
- Mutually exclusive $\neq$ independent.
- $P(A|B) + P(A|B^c) \neq 1$. Correct: $P(A|B) + P(A^c|B) = 1$.
- $E[X^2] \neq (E[X])^2$ unless $\mathrm{Var} = 0$.
- Geometric starts at $k = 1$; NegBin(r,p) at $k = r$.
- When probability depends on a random N: **must condition** on N using LOTP.
- Always **justify** steps: name distributions, cite formulas, state independence.
- Don't forget to divide by P(conditioning event).